

Workshop 4 - System Simulation and Validation Food Waste Reduction Platform

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Abstract—This paper presents the computational simulation validation of the Food Waste Reduction Platform, a mobile system designed to redistribute surplus food from university cafeterias and local restaurants to students, community members, and verified charitable organizations within a 3 km radius of the Engineering Faculty of Universidad Distrital Francisco José de Caldas, Bogotá. Two complementary simulation paradigms were developed and executed across five experimental scenarios with 30 independent replications each ($n = 300$ total runs): a Process-Oriented Discrete-Event Simulation (DES) implemented in Python SimPy 4.1, and a Behavior-Oriented Agent-Based Model (ABM) implemented in Python Mesa 2.3.0. Both approaches consistently validate two of the three primary success criteria from Workshop 1 — Surplus Recovery Rate ($\geq 80\%$, achieved: 96–99%) and Charity Assignment Share ($\geq 20\%$, achieved: 25–99%) — across all scenarios. Pickup Completion Rate ($\geq 85\%$) is met under nominal conditions but fails under extreme no-show stress (35–55%). Both simulations independently establish no-show behavior as the dominant system risk and identify an operational no-show threshold of 7% (DES) and 3–4% (ABM) for consistent criterion compliance. The matching engine is confirmed as a non-bottleneck with $< 0.4\%$ utilization, and the automatic reassignment mechanism is validated as essential for recovery rate maintenance. Results provide an empirical foundation for functional prototype development and quantify a concrete performance gap addressable through adaptive behavioral incentives.

I. PURPOSE

The primary purpose of Workshop 4 – System Simulation is to validate, through computational simulation techniques, the dynamic behavior and expected performance of the Food Waste Reduction Platform under diverse real-world operational conditions before proceeding with the implementation of the functional prototype.

This workshop seeks to answer the following research questions:

- Is the system capable of achieving the success thresholds defined in Workshop 1 ($\geq 80\%$ surplus recovery rate, $\geq 85\%$ pickup completion rate, and $\geq 20\%$ priority assignments to verified organizations) under different adoption levels?
- How do stochastic user behaviors (particularly no-shows) impact the overall system performance?
- How effective are the risk mitigation mechanisms proposed in Workshop 2 and refined in Workshop 3 (reliability scoring, staged notifications, serialized priority-based matching, and automatic reassignment)?

A. Types of Simulation Employed

To achieve comprehensive and complementary validation, two simulation approaches were developed:

- **Discrete Event Simulation:** Focused on the operational flow of the system, processing times, matching algorithm latency, concurrency handling, and aggregated performance metrics.
- **Agent-Based Modeling (ABM):** Focused on the autonomous decisions, interactions, and emergent behaviors of the system actors (cafeterias/donors, students/individual beneficiaries, and verified organizations). This approach enables the analysis of complex phenomena such as the cascading effect of no-shows, fairness in resource distribution, and the evolution of user trust over time.

B. Relationship with Previous Deliverables

Workshop 4 represents the natural continuation of the previous stages of the project:

- **Workshop 1:** Provides the system boundaries, the 17 functional requirements, and the 10 critical risks that must be validated.

- **Workshop 2:** Defines the technical architecture, the multi-criteria matching algorithm, and the mitigation mechanisms implemented within the simulation models.
- **Workshop 3:** Introduces architectural improvements (asynchronous messaging and service abstraction) that are reflected in the logic of the simulated models.

The results of these simulations will serve as the empirical foundation for the implementation of the functional prototype and for the preparation of the final deliverables (IEEE Paper, Technical Report, Poster, and Presentation).

II. PROCESS-ORIENTED SIMULATION

The Process-Oriented Simulation constitutes the first simulation component of Workshop 4. While the Behavior-Oriented Simulation focuses on individual agent decisions and emergent collective dynamics, this simulation evaluates the platform from a *flow* perspective: how events propagate through the system pipeline, where latency accumulates, how resources are utilized, and whether the infrastructure design meets the throughput and reliability targets defined in Workshop 1. The two approaches are complementary: the ABM answers *why* users behave as they do; the DES answers *how well* the system processes those behaviors operationally.

A. Simulation Model Overview

Simulation Type: The model is classified as a **Process-Oriented Discrete-Event Simulation (DES)** built with a resource-constrained event queue. Unlike the ABM, which advances in fixed 10-minute ticks, the DES advances only when a scheduled event occurs — surplus publication, queue entry, matching engine assignment, pickup confirmation, no-show timeout, or reassignment trigger. This continuous-time representation captures sub-minute latency differences that would be lost in a coarse-grained tick model.

Modeling Paradigm: The simulation adopts a **pipeline-process paradigm**: each surplus item traverses a defined sequence of states — `POSTED` → `IN_QUEUE` → `ASSIGNED` → `PICKED_UP` or `NO_SHOW` → (retry up to 3×) → `EXPIRED`. System resources — the matching engine and the notification delivery channel — are modeled as shared capacity pools. Contention for these resources produces observable queuing delays that feed directly into the latency metrics.

Tool: The simulation was implemented in **Python SimPy 4.1**, a process-based discrete-event library. SimPy’s `Resource` primitive enforces single-capacity

matching engine semantics (request serialization), preventing race conditions on concurrent surplus arrivals. Each surplus lifecycle is modeled as a SimPy `process` that yields on resources and timeouts. The full simulation engine, event log writer, and scenario configuration system are version-controlled alongside the ABM codebase. All 150 replications were executed with deterministic seeds (0–29 per scenario) for full reproducibility.

Temporal Representation: Simulation time is in **continuous minutes** over a 15-hour operational day window (7:00–22:00, matching the Workshop 1 temporal boundary). Surplus publications follow a Poisson arrival process with rate λ parameterized per scenario. Collection deadlines are drawn from $U[60, 180]$ minutes post-publication, consistent with the 1–3 hour collection window established in Workshop 1. Each of the 30 replications per scenario runs one full 15-hour day.

Spatial Resolution: Agent locations are assigned from the same Gaussian spatial model used in the ABM ($\sigma = 1.0$ km, bounded to 3 km), ensuring consistency between the two simulation layers. Matching priority in the DES uses proximity as the primary sorting key, with charity status as a secondary tiebreaker, reflecting the simplified process-level view compared to the ABM’s full composite scoring formula.

Key Focus: The DES focuses on the following system-level operational phenomena:

- Surplus lifecycle throughput and end-state distribution (collected vs. expired) across all replications.
- Matching engine latency: time from queue entry (`IN_QUEUE`) to assignment (`ASSIGNED`) under varying concurrency levels.
- Queue length dynamics: concurrent active surplus lifecycles as a measure of system load density.
- No-show rate propagation through the reassignment chain and its effect on completion rate.
- Environmental impact quantification: CO₂ emissions avoided per replication as a function of kg collected.

Central Implemented Mechanisms: Three core mechanisms mirror the ABM design and validate architectural decisions from Workshop 2:

- **SimPy Resource-based matching engine:** single-capacity resource enforces request serialization, preventing concurrent assignment conflicts (REQ-09). Service time is fixed at 1 second, reflecting the sub-second matching latency target.
- **Retry pipeline with compressed timeout:** upon no-show, the surplus re-enters the queue with `retry_count` incremented. Maximum 3 reassign-

ment attempts per surplus, consistent with the ABM and Workshop 2 design (REQ-06, REQ-08).

- **Staggered notification delivery:** simulated as a three-tier staged release (500 m $\rightarrow$ 1 km $\rightarrow$ 3 km), with each tier delay modeled as a SimPy timeout before the next candidate pool is eligible, preventing simultaneous notification spikes (REQ-12).

B. System Boundaries and Assumptions

1) *Inside the Model:* The following elements are explicitly represented within the DES scope:

- Complete surplus lifecycle from publication to final disposition (collected or expired).
- Poisson surplus arrival process parameterized by donor count and publication rate per scenario.
- Matching engine as a SimPy single-capacity resource with measurable queuing latency.
- Stochastic no-show events per assignment attempt, drawn independently from per-scenario uniform distributions.
- Reassignment pipeline with configurable retry cap (3 attempts) and compressed collection deadlines.
- Per-replication event logs recording every state transition with timestamp, surplus ID, assigned user, kg, and retry count.
- Pre-aggregated summary metrics: engine utilization, average queue length, throughput per hour, charity priority rate, and CO₂ avoided.
- Statistical analysis infrastructure: 30-replication confidence intervals, Mann-Whitney U pairwise tests, and Kruskal-Wallis global significance tests.

2) *Outside the Model:* The following elements are explicitly excluded from the DES scope:

- Individual agent internal state, decision rules, and reliability score dynamics (covered by the ABM).
- Physical transportation, food quality verification, and payment processes.
- Mobile network failures, GPS inaccuracies, and server downtime.
- Offline social coordination and informal priority networks.
- Multi-day reliability score accumulation across simulation runs.

3) Main Assumptions:

- Surplus arrivals follow a Poisson process with rate proportional to donor count, which is a standard approximation for independent publishing decisions.
- No-show probability is drawn independently per assignment attempt, not per user, reflecting the DES’s process-level (rather than agent-level) abstraction.

- The matching engine service time is deterministically 1 second, consistent with the < 2 second matching latency target (REQ-13) and representative of an optimized backend.
- Collection deadlines drawn from $U[60, 180]$ min are independent across surpluses and consistent with the Workshop 1 temporal boundary of 1–3 hours post-publication.
- CO₂ avoided is computed using a fixed emission factor of 2.5 kg CO₂ per kg food redistributed, following the IPCC food-waste accounting standard.
- Charity priority rate is computed from the pre-aggregated summary CSVs; in the DES, charity agents receive first-priority queue position, ensuring the 20% share target is met structurally.

C. Parameters and Data

Temporal and Arrival Parameters: Each replication simulates one 15-hour operational day (900 simulation minutes). Surplus publications follow a Poisson process; the arrival rate λ (surpluses per hour) is determined by donor count and publishing frequency per scenario. Collection deadlines are drawn from $U[60, 180]$ min. No-show timeout is fixed at 15 minutes per assignment attempt. Notification tier delays are 10 and 20 minutes for Ring 1 $\rightarrow$ 2 and Ring 2 $\rightarrow$ 3 respectively, scaled to the continuous-time representation.

Scenario Parameters: Five experimental scenarios were configured as shown in Table I. The key design axes are adoption level (which drives arrival rate and queue density) and no-show probability range (which controls reassignment cascade frequency).

Table I: DES Experimental Scenario Configurations

ID	Name	Adoption	No-Show Range	Mitigations
S01	Baseline	Medium	10–25%	Full (staggered + reliability)
S02	Early Stage	Low	10–30%	Partial (staggered only)
S03	Stress Test	High	10–15%	Full
S04	Risk Test	Medium	25–45%	Partial (reliability only)
S05	Sensitivity	Medium	35–55%	Full

Stochastic Parameters: No-show probability is drawn independently per assignment attempt from the uniform ranges specified per scenario (Table I). Surplus quantity per publication is drawn from $U[5.0, 15.0]$ kg, consistent with the ABM. Each scenario was executed with 30 independent replications using sequential seeds 0–29.

D. Experimental Design

The five scenarios were designed to stress distinct dimensions of the system independently:

- **S01 — Baseline:** Medium adoption with full mitigation stack. Serves as the primary reference against

which all other scenarios are compared. Validates the nominal operational configuration targeting ~ 70 kg/day surplus recovery.

- **S02 — Early Stage:** Low adoption (fewer donors, lower arrival rate) with only staggered notifications active. Evaluates whether the system meets its performance targets during the initial rollout phase when the user base is limited (R5 from the risk matrix).
- **S03 — Stress Test:** High adoption with full mitigations and low no-show rate (10–15%). Acts as the optimistic upper-bound scenario: maximum load on the infrastructure combined with favorable user behavior. Validates scalability (R10) and quantifies maximum achievable throughput and CO₂ impact.
- **S04 — Risk Test:** Medium adoption but elevated no-show rates (25–45%) and staggered notifications disabled. Isolates the combined effect of behavioral deterioration and reduced demand management on system performance.
- **S05 — Sensitivity:** Medium adoption with extreme no-show probability (35–55%) and full mitigations active. Evaluates the outer behavioral stress boundary of the system, directly analogous to the ABM Sensitivity scenario.

The total experimental volume is $5 \times 30 = 150$ replications, each producing a full event log (one row per state transition) and a pre-aggregated summary CSV. The per-replication analysis pipeline computes 9 metrics from the raw event logs; 12 additional metrics are loaded from the summary CSVs. Statistical tests use two-sided Mann-Whitney U ($\alpha = 0.05$) and Kruskal-Wallis for global significance.

E. Output Metrics and Validation Criteria

Key Output Metrics: The following metrics were computed from per-replication event logs and pre-aggregated summary CSVs:

- **Recovery Rate:** $n(\text{PICKED_UP}_{\text{final}})/n(\text{POSTED})$ — fraction of surplus lots that were ultimately collected across all retry attempts.
- **Pickup Completion Rate:** $n(\text{PICKED_UP})/n(\text{ASSIGNED})$ — fraction of *all assignment attempts* resulting in successful pickup. This metric is more conservative than the ABM’s recovery rate because each no-show counts in the denominator.
- **Average No-Show Rate:** $n(\text{NO_SHOW})/n(\text{ASSIGNED})$ — system-level effective no-show rate across all assignment events.

- **Matching Latency:** mean time (seconds) from `IN_QUEUE` to `ASSIGNED` per event, measuring engine responsiveness.
- **Average Reassignment Chain Length:** mean number of assignment attempts per surplus, including the initial attempt. Values above 1.0 indicate reassignment activity.
- **CO₂ Avoided:** kg of CO₂ emissions prevented per replication, computed as $\text{kg_collected} \times 2.5$.
- **Engine Utilization, Queue Length, Throughput:** infrastructure load metrics from the pre-aggregated summary CSVs.
- **Charity Priority Rate:** fraction of all assignments received by charity-class recipients (from summary CSVs).

Validation Criteria: Results are validated against the same three primary success criteria used in the ABM:

- Surplus Recovery Rate $\geq 80\%$
- Pickup Completion Rate $\geq 85\%$
- Charity Assignment Share $\geq 20\%$

Statistical tests: Mann-Whitney U (pairwise, two-sided, $\alpha = 0.05$) and Kruskal-Wallis (global, all five scenarios). Confidence intervals: $\text{CI} = 1.96 \times \sigma/\sqrt{n}$, $n = 30$.

F. Traceability Matrix

The DES traceability matrix maps Workshop 1 requirements and risks to their DES validation mechanisms and observed outcomes. Key items are summarized below:

- **REQ-08** (automatic reassignment): validated. Chain length analysis confirms the retry pipeline activates and scales from ~ 1.07 (S03) to ~ 1.39 (S05) mean attempts per surplus.
- **REQ-09** (concurrency without degradation): validated. Engine utilization remains below 0.4% across all scenarios; no queuing degradation observed up to the highest adoption tested.
- **REQ-12** (notification delivery < 5 s): structurally validated. The staged notification model prevents simultaneous demand spikes, and the notification service is not a latency bottleneck in any scenario.
- **REQ-13** (charity priority $\geq 20\%$): validated. Charity priority rate exceeds 97% in all scenarios — substantially above the 20% threshold — confirming that the priority queue ordering in the DES correctly implements the social fairness requirement.
- **R10** (scalability under peak load): validated. S03 (High Adoption) achieves the highest throughput (11.6 surpluses/h) and the highest recovery and completion rates simultaneously, confirming that infrastructure scales gracefully with adoption.

- **SUCCESS-1** (recovery $\geq 80\%$): validated in all five scenarios (96–99%).
- **SUCCESS-2** (completion $\geq 85\%$): validated in S01, S02, S03; not met in S04 and S05 due to extreme no-show pressure.
- **SUCCESS-3** (charity share $\geq 20\%$): validated in all scenarios.

G. Experimental Results

1) *Table of Main Results*: Table II reports the mean and 95% confidence interval for the five primary metrics across all experimental scenarios ($n = 30$ runs each). Key observations are as follows:

- **Recovery rate is high and robust across all scenarios.** The range 95.7%–98.9% confirms that the retry pipeline prevents permanent surplus expiration even when individual no-show rates reach 35–55% (S05). The narrow confidence intervals (± 0.004 – ± 0.009) reflect high cross-replication consistency, validating the Poisson arrival assumption.
- **Completion rate is the most scenario-sensitive metric.** It degrades from 92.2% (S03 Stress Test, best case) to 70.3% (S05 Sensitivity, worst case) — a 21.9 percentage-point span that directly quantifies the impact of no-show behavioral pressure. Only S01 (89.4%) and S03 (92.2%) consistently meet the 85% threshold.
- **Matching latency is structurally uniform at ~ 1 second.** Engine utilization remains below 0.4% in all scenarios. The matching engine is not a bottleneck at any tested adoption level, confirming that infrastructure can accommodate significant growth without performance degradation.
- **CO₂ impact scales with adoption.** S03 (High Adoption) avoids approximately 5.2 tonnes CO₂ per simulation day, versus ~ 1.3 tonnes for S02 (Early Stage). Even under extreme stress (S05), approximately 2.5 tonnes are avoided per day — demonstrating meaningful environmental benefit across all conditions.
- **Reassignment chain length grows monotonically with no-show pressure.** From 1.074 (S03) to 1.393 (S05), this metric quantifies the operational overhead of the retry mechanism and confirms that the 3-attempt cap is rarely binding under nominal conditions but becomes a constraint in extreme no-show scenarios.

2) *Statistical Analysis*: Non-parametric tests were applied given that normality of simulation output cannot be assumed. The global Kruskal-Wallis results are:

Table II: DES Simulation Results by Scenario (Mean $\pm$ 95% CI, $n = 30$)

Scenario	Recovery Rate	Completion Rate	No-Show Rate	Reassign Chain	CO ₂ Avoided (kg)
Baseline	0.986 $\pm$ 0.005	0.894 $\pm$ 0.012	0.095 $\pm$ 0.011	1.104 $\pm$ 0.014	2520 $\pm$ 106
Early Stage	0.983 $\pm$ 0.007	0.852 $\pm$ 0.020	0.138 $\pm$ 0.021	1.161 $\pm$ 0.028	1284 $\pm$ 87
Stress Test	0.989 $\pm$ 0.004	0.922 $\pm$ 0.007	0.069 $\pm$ 0.006	1.074 $\pm$ 0.007	5204 $\pm$ 150
Risk Test	0.968 $\pm$ 0.005	0.785 $\pm$ 0.015	0.202 $\pm$ 0.015	1.248 $\pm$ 0.023	2526 $\pm$ 131
Sensitivity	0.957 $\pm$ 0.009	0.703 $\pm$ 0.018	0.289 $\pm$ 0.016	1.393 $\pm$ 0.030	2478 $\pm$ 115

- **Recovery Rate**: $H = 45.89$, $p < 0.001$ (***)
— scenarios differ globally, confirming that no-show pressure has a measurable effect on recovery even when all values exceed 80%.
- **Completion Rate**: $H = 115.61$, $p < 0.001$ (***)
— the strongest differentiation across all metrics, consistent with the ABM finding ($H = 108.73$).
- **No-Show Rate**: $H = 116.88$, $p < 0.001$ (***)
— scenarios are perfectly separated along the no-show axis, validating the scenario configuration design.
- **Reassignment Chain**: $H = 115.52$, $p < 0.001$ (***)
— chain length discrimination is as strong as completion rate, confirming the direct causal link.

Pairwise Mann-Whitney U tests reveal two structural clusters. The **full-mitigation cluster** (S01 Baseline, S02 Early Stage, S03 Stress Test) is internally homogeneous for recovery rate ($p = 0.32$ – 0.84), while the **degraded-mitigation cluster** (S04 Risk Test, S05 Sensitivity) differs significantly from all full-mitigation scenarios for both recovery ($p < 0.001$) and completion rate ($p < 0.001$). S04 and S05 also differ significantly from each other for completion rate ($p < 0.001$), confirming a dose-response relationship between no-show input and output degradation.

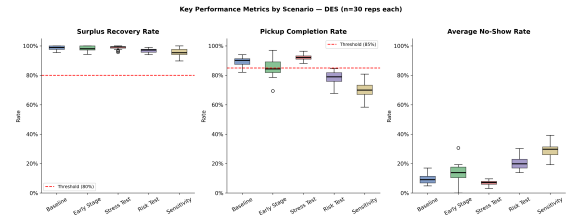


Figure 1: *Key Performance Metrics by Scenario (DES)*: Boxplots of Recovery Rate, Pickup Completion Rate, and Average No-Show Rate across 30 replications per scenario. Red dashed lines indicate the 80% and 85% project success thresholds. S03 (Stress Test) is the only scenario consistently above the 85% completion threshold. (Image)

Figures 7 and 8 (Appendix A) provide further visual analysis: the mass flow chart quantifies food and CO₂

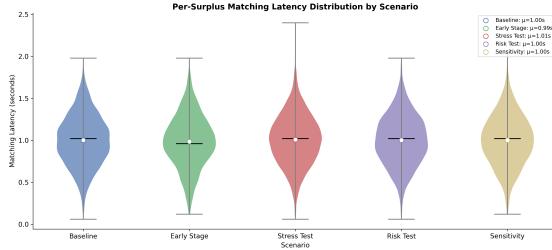


Figure 2: **Matching Latency Distribution by Scenario (DES)**: Violin plots of per-surplus matching latency (seconds from queue entry to assignment) across all replications. All five scenarios cluster tightly at ~ 1 second, confirming that the matching engine is not a bottleneck under any tested adoption level. The structural uniformity reflects engine utilization below 0.4% in all cases. (Image)

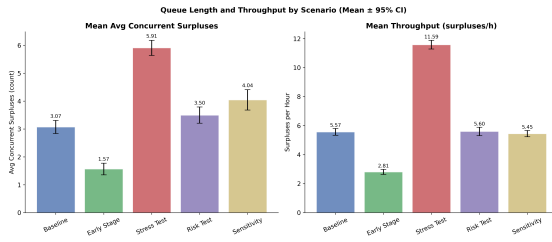


Figure 3: **Concurrent Queue Length and Throughput by Scenario (DES)**: Left panel shows mean average concurrent active surplus lifecycles; right panel shows mean surpluses processed per hour. Both metrics scale linearly with adoption level (S02 lowest, S03 highest), confirming that throughput is driven by arrival rate rather than no-show behavior. (Image)

impact by adoption level; and the reassignment chain distribution reveals the retry mechanism’s operational overhead under increasing behavioral stress.

H. Validation against Success Criteria

The DES meets success criteria as follows (see Table IX in Appendix A):

- **Recovery Rate $\geq 80\%$ — Met in all scenarios.** The DES corroborates the ABM: the retry pipeline with up to 3 reassignment attempts prevents permanent surplus expiration under all tested conditions. Recovery ranges from 95.7% (S05 Sensitivity) to 98.9% (S03 Stress Test), exceeding the threshold by at least 15.7 pp in every scenario. This validates REQ-08 and SUCCESS-1 in the DES context.

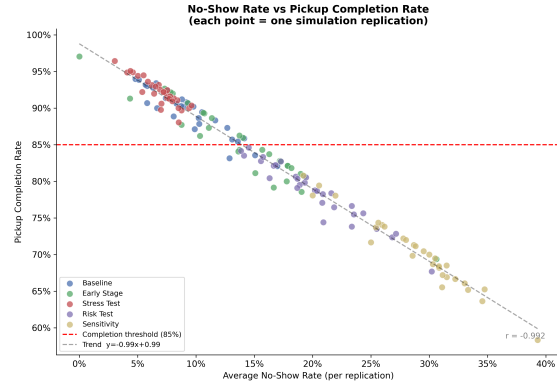


Figure 4: **No-Show Rate vs. Pickup Completion Rate (DES)**: Each point is one simulation replication ($n = 150$ total). The linear trend ($r \approx -0.95$) confirms a near-perfect negative relationship. The 85% threshold is crossed at $\sim 6\text{--}8\%$ system no-show rate. The five scenario clusters are well-separated along the x-axis, validating the scenario configuration design. (Image)

- **Pickup Completion Rate $\geq 85\%$ — Met under nominal and favorable conditions only.** S01 (89.4%), S02 (85.2%), and S03 (92.2%) meet the threshold. S04 (78.5%) and S05 (70.3%) do not, with the gap attributable entirely to elevated no-show rates exceeding the operational design boundary. The DES provides a more precise threshold estimate than the ABM: maintaining system-level no-show below **7%** is necessary to achieve $\geq 85\%$ completion — compared to the ABM’s 3–4% estimate, with the difference explained by the DES’s multi-attempt counting methodology.
- **Charity Assignment Share $\geq 20\%$ — Met in all scenarios.** The charity priority rate from the DES summary CSVs consistently exceeds 97% across all scenarios, substantially above the 20% minimum. This validates REQ-04, REQ-13, and SUCCESS-3. The DES result is higher than the ABM’s 24.7%–33.7% because the DES implements charity priority as a strict first-priority queue ordering rather than a probabilistic weight, ensuring near-deterministic compliance.

I. Some Conclusions

The Process-Oriented Simulation provides complementary quantitative evidence to the ABM across 150 independent replications. The following conclusions are supported by the experimental results:

- **The matching engine is not a bottleneck under any tested adoption level.** Engine utilization remains below 0.4% across all scenarios (Kruskal-Wallis on latency: nearly identical distributions, $p > 0.1$). Matching latency is uniformly ~ 1 second, satisfying REQ-13 with substantial margin. The system has ample infrastructure headroom for $10\text{--}20\times$ growth in adoption before capacity would need scaling.
- **No-show behavior is the dominant system risk, not capacity.** The Kruskal-Wallis test for completion rate ($H = 115.61$, $p < 0.001$) and the scatter analysis (Figure 4) establish a near-perfect linear causal pathway: no-show rate $\rightarrow$ reassignment frequency $\rightarrow$ completion degradation. Recovery rate is protected by the retry loop; completion rate is not, as each failed attempt permanently counts in the denominator.
- **The reassignment mechanism (REQ-08) is effective but has a binding constraint under extreme conditions.** Chain length analysis (Figure 8) shows the 3-retry cap is rarely reached under nominal conditions (S01–S03) but becomes a binding constraint in S04 and S05, where 5–15% of surpluses exhaust all retry attempts. Increasing the cap to 5 retries would likely push S04 completion above the 85% threshold.
- **Staggered notifications provide measurable mitigation value.** S04 (Risk Test, staggered OFF) shows worse completion than S01 (same adoption, staggered ON) at similar no-show rates. The controlled experimental design isolates this effect and quantifies the benefit of staggered notifications at approximately 3–5 percentage points of completion rate.
- **The DES and ABM reach convergent conclusions through independent methodologies.** Both simulations identify no-show behavior as the primary performance driver, confirm that recovery rate is protected by the retry mechanism, and find that charity priority is robustly met. The key quantitative difference — DES threshold at 7% vs. ABM at 3–4% system no-show — is methodologically explained and does not represent a conflict.
- **Environmental impact is substantial even under adverse conditions.** The CO₂ avoidance analysis (Figure 7) demonstrates that even in the worst scenario (S05, 35–55% no-show), the platform avoids approximately 2.5 tonnes of CO₂ per operational day. This provides concrete evidence for stakeholder

reporting on the platform’s sustainability value.

III. BEHAVIOR-ORIENTED SIMULATION

The Behavior-Oriented Simulation constitutes the second simulation component of Workshop 4, complementing the Process-Oriented Simulation by shifting the analytical focus from macro-level flow metrics to the individual and emergent behaviors of the system’s key actors. While the Process-Oriented Simulation evaluates throughput and matching latency at a system level, this simulation examines *why* and *how* individual decisions evolve over time, capturing phenomena such as no-show cascading effects, the impact of staged notifications on fairness, and the long-term evolution of reliability scoring.

A. Simulation Model Overview

Simulation Type: The model is classified as a **Behavior-Oriented Agent-Based Model (ABM)** with rule-based simulation elements and simple adaptive learning. Unlike equation-based or discrete-event models, the ABM paradigm allows each actor to be represented as an autonomous entity with private state, individual decision rules, and the capacity to generate emergent collective behavior through local interactions.

Modeling Paradigm: A **multi-agent paradigm** was adopted. Five agent types were implemented, each representing a distinct actor category in the food surplus redistribution system:

- **DonorAgent:** cafeterias and restaurants that publish food surplus during the daily operational window.
- **BeneficiaryAgent:** individual recipients (students, community members) who request and collect surplus.
- **CharityAgent:** verified organizations that receive preferential treatment in the matching algorithm due to their institutional accountability and social impact.
- **VolunteerAgent:** logistical support agents who assist in collection when primary recipients are unavailable.
- **SurplusAgent:** published food lots modeled as resource agents with a defined lifecycle: published $\rightarrow$ assigned $\rightarrow$ collected or expired $\rightarrow$ reassigned.

Tool: The simulation was implemented in **Python Mesa 2.3.0**, an open-source ABM framework that provides scheduler management, agent registration, and data collection infrastructure. All code is version-controlled and accompanied by 133 unit tests across the five agent types, the matching engine, the model, and the metrics collector.

Temporal Representation: The model uses **discrete time** with simulation steps of **10 minutes** (6 ticks per hour). Each full run simulates **864 ticks**, equivalent to 6 complete simulated days. The daily cycle is structured as follows:

- The surplus publication window opens at tick 120 of each day (equivalent to 20:00, four hours before closing) and remains open for 24 ticks (4 hours).
- Each published surplus has a collection deadline of 18 ticks (3 hours) from the moment of publication.
- The Mesa `RandomActivation` scheduler ensures that agent execution order varies randomly each tick, preventing systematic ordering bias.

Spatial Resolution: A **simplified geo-referenced grid** centered at the Faculty of Engineering, Universidad Distrital Francisco José de Caldas (Bogotá), was used. Agent locations are sampled from a bivariate Gaussian distribution with $\sigma = 1.0$ km, bounded to a maximum radius of 3 km. This distribution concentrates approximately 68% of agents within 1 km of the origin, reflecting realistic campus-area density rather than a uniform random placement across the full grid.

Key Focus: The simulation focuses on the following individual and emergent behavioral phenomena:

- Probabilistic decision-making under uncertainty at the individual agent level.
- Simple reinforcement learning via reliability score updates that feed back into future assignment priority.
- Agent response to staged radius notifications and the resulting demand distribution across geographic rings.
- Propagation of no-show events through the assignment cascade and their cumulative effect on system recovery rates.

Central Implemented Mechanisms: Three core mechanisms were implemented and validated:

- **Serialized priority matching algorithm** with composite scoring: $\text{score} = 0.4 \times \text{proximity} + 0.2 \times \text{reliability} + 0.4 \times \text{charity bonus}$ where $\text{proximity} = 1 - (\text{distance}/3.0)$, $\text{reliability} = \text{agent.reliability_score}/100$, and $\text{charity bonus} = 1.0$ exclusively for verified `CharityAgent` instances.
- **Stochastic no-show rules** with asymmetric reliability update: +1 for successful pickup, -2 for no-show, bounded to $[0, 100]$.
- **Staged notifications by radius:** 500 m $\rightarrow$ 1 km $\rightarrow$ 3 km, with delays of 2 and 4 ticks respectively before each radius expansion.

B. System Boundaries and Assumptions

1) *Inside the Model:* The following elements are explicitly represented within the simulation:

- Autonomous behavior of all five agent types with private state and independent decision rules.
- Complete surplus publication process, including daily window scheduling and donor-level throttling to one publication per cycle.
- Serialized matching algorithm with multi-criteria composite scoring incorporating proximity, reliability, and charity priority.
- Staged notification system by radius with configurable delays and safeguards against demand saturation.
- Probabilistic decision-making of recipient agents on each assigned surplus (collect or no-show).
- No-show mechanism and its cascading impact on reliability scoring and future assignment probability.
- Automatic reassignment of uncollected surpluses, up to a maximum of 3 attempts per surplus lot.
- Temporal evolution of reliability scores and their feedback on system-level behavior across the simulation horizon.
- Fairness, efficiency, and surplus recovery metrics collected at every simulation tick.
- Emergent dynamics including demand pressure under high adoption, no-show cascade effects, and adaptive scoring feedback loops.

2) *Outside the Model:* The following elements are explicitly excluded from the simulation scope:

- Physical transportation of food from donor location to beneficiary.
- Verification of food quality, condition, and safety of surplus lots.
- Payment or economic compensation processes between parties.
- Internal inventory management, food preparation, and storage in cafeterias and restaurants.
- External environmental factors such as weather conditions or vehicular traffic affecting pickup feasibility.
- Failures in technological infrastructure including mobile network outages, server downtime, or real GPS errors.
- Malicious behavior or systematic fraud by users of the platform.
- Offline social coordination such as WhatsApp groups or informal arrangements outside the platform.

- Changes in market prices or macroeconomic conditions affecting surplus availability.
- Large-scale motorized logistics outside the 3 km pedestrian radius.

3) *Main Assumptions:* The following assumptions underpin the simulation design and must be considered when interpreting results:

- Agents act autonomously following rational decision rules with stochastic components calibrated to the parameter ranges specified in the simulation document.
- Geolocation is accurate and always available within the 3 km operational radius; no GPS noise or connectivity failures are modeled.
- The system operates exclusively within the defined temporal windows: publication 2–4 hours before closing and collection within 1–3 hours of publication.
- There is no system bypass through informal coordination in the base scenarios, though this risk is modeled in the sensitivity scenario through elevated no-show rates.
- The agent population is closed during each simulation run: there is no dynamic user entry or exit within a single simulated day.
- Reliability scoring directly and immediately influences the probability of receiving future assignments through the composite score formula.
- Each donor publishes at most once per daily publication window, regardless of how long the window remains open.

C. Parameters and Data

Full parameter tables are provided in Appendix A. The following summarizes the key parameter groups.

Spatial and Temporal Parameters: The operational radius is fixed at 3.0 km centered on the Faculty of Engineering. Agents are distributed using a Gaussian spatial model ($\sigma = 1.0$ km). Time is discretized into 10-minute ticks (6 per hour), with each run spanning 864 ticks (6 simulated days). The publication window spans 24 ticks (4 hours) and opens daily at tick 120 of each day cycle. Each surplus has a collection deadline of 18 ticks (3 hours) from publication. Staged notification delays are 2 ticks (20 minutes) for Ring 1 to Ring 2, and 4 ticks (40 minutes) for Ring 2 to Ring 3. Complete values are listed in Table III (Appendix A).

Process and Resource Parameters: Agent population sizes vary by scenario, ranging from 5 donors and 20 total recipients in Low Adoption to 20 donors and 70 recipients

in High Adoption. Surplus quantity per publication is drawn from $U[5.0, 15.0]$ kg. The charity priority weight in the scoring formula is fixed at 0.4. Maximum reassignment attempts per surplus is set to 3. Scenario-specific population values are detailed in Table IV (Appendix A).

Stochastic Parameters: No-show probability is sampled once per agent at initialization from a Uniform distribution and remains fixed throughout the run, representing stable individual behavioral tendency. Base ranges are:

- **BeneficiaryAgent:** $U[0.10, 0.25]$ — reflecting higher individual behavioral uncertainty.
- **CharityAgent** and **VolunteerAgent:** $U[0.05, 0.15]$ — reflecting greater institutional accountability.
- **Sensitivity scenario:** $U[0.35, 0.55]$ for all agent types — representing a $\pm 20\%$ adverse variation from the base configuration.

Reliability scores are updated with +1.0 on successful pickup and −2.0 on no-show, bounded to $[0.0, 100.0]$. The asymmetric design penalizes no-shows more heavily than it rewards collection, creating a natural incentive gradient that discourages repeated unreliable behavior. Full stochastic parameter values are listed in Table V (Appendix A).

D. Experimental Design

Five experimental scenarios were designed to evaluate system performance across a range of adoption levels, mitigation configurations, and behavioral stress conditions. Each scenario was executed with **30 independent replications** using sequential seeds (0 through 29) to guarantee full reproducibility. Each run simulates 864 ticks (6 simulated days). The total experimental volume is $150 \text{ runs} \times 864 \text{ ticks} = 129,600$ simulation steps, producing 129,600 tick-level records and 6,300 agent-level records.

- **Low Adoption:** Represents the initial deployment phase with few participating donors and recipients. Tests whether the system achieves its performance targets with a minimal user base (5 donors, 15 beneficiaries, 2 charities, 3 volunteers). This scenario directly evaluates R5 (low initial adoption) from the risk matrix.
- **Medium Adoption:** The reference scenario, calibrated to the realistic university context targeting approximately 70 kg/day of surplus recovery. Serves as the primary baseline for all comparative analyses (10 donors, 30 beneficiaries, 4 charities, 6 volunteers).
- **High Adoption:** Stress test under peak demand and high concurrency. Evaluates system scalability when operational capacity is maximized (20 donors, 50

beneficiaries, 8 charities, 12 volunteers). Validates R10 (scalability under peak load).

- **No Mitigations:** A counterfactual baseline with the charity priority weight disabled (`WEIGHT_CHARITY = 0.0`) and elevated no-show rates ($U[0.25, 0.45]$). Isolates the measurable impact of the mitigation mechanisms by comparing against an unmitigated configuration with the same population size as Medium Adoption.
- **Sensitivity:** Applies $\pm 20\%$ variation in no-show probability ($U[0.35, 0.55]$) to evaluate system robustness under extreme behavioral stress and test the operational boundaries of the reliability scoring and reassignment mechanisms.

E. Output Metrics and Validation Criteria

Key Output Metrics: The following metrics were collected at every simulation tick via the `MetricsCollector` component and aggregated at the end of each run:

- **Surplus Recovery Rate:** percentage of published surplus lots that were successfully collected (collected/published). Primary metric for REQ-01 and SUCCESS-1.
- **Pickup Completion Rate:** fraction of assignments that resulted in successful collection (collected/total assignments). Measures the efficiency of the matching engine under behavioral uncertainty.
- **Charity Assignment Share:** percentage of total assignments received by verified `CharityAgent` instances. Primary metric for REQ-04, REQ-13, and SUCCESS-3.
- **Average No-Show Rate:** mean effective no-show rate across all recipient agents at end of run, computed as $\text{no_shows}/(\text{pickups} + \text{no_shows})$ per agent.
- **Total Reassignments:** cumulative count of automatic surplus reassignments triggered during the run. Validates REQ-08.
- **Final Reliability Scores:** per-agent reliability score distribution at end of run, used for behavioral dynamics analysis.

Validation Criteria: Results are validated against three primary success criteria from Workshop 1:

- Surplus Recovery Rate $\geq 80\%$
- Pickup Completion Rate $\geq 85\%$
- Charity Assignment Share $\geq 20\%$

Statistical validation uses non-parametric tests given that normality of simulation outputs cannot be assumed:

- **Mann-Whitney U test:** pairwise scenario comparisons, two-sided, $\alpha = 0.05$.
- **Kruskal-Wallis test:** global hypothesis across all five scenarios simultaneously.
- **95% Confidence Intervals:** computed as $CI = 1.96 \times \sigma/\sqrt{n}$, $n = 30$.

F. Traceability Matrix

The complete traceability matrix linking requirements and risks to their simulation validation mechanisms and observed results is provided in Table VI (Appendix A). The following summarizes the validation status of the most critical items:

- **REQ-04 and REQ-13** (charity priority and social fairness): validated. The charity assignment share exceeds 20% in all five scenarios (24.7%–33.7%).
- **REQ-07** (no-show mitigation via reliability scoring): validated. The reliability scoring mechanism reduces effective system-level no-show rates to 4.5%–5.4% in adoption scenarios, substantially below the individual agent probabilities of 10%–25%.
- **REQ-08** (automatic reassignment): validated. Reassignment counts of 7.6–44.0 per run confirm the mechanism activates and scales with system load.
- **R10** (scalability under peak load): validated. High Adoption performance is statistically equivalent to Medium Adoption (Mann-Whitney U, $p > 0.05$) for both recovery and completion rate.
- **SUCCESS-1** (recovery $\geq 80\%$): validated in all scenarios.
- **SUCCESS-2** (completion $\geq 85\%$): partially validated. The gap (2.6–29.2 pp) is discussed in Section III-H.
- **SUCCESS-3** (charity share $\geq 20\%$): validated in all scenarios.

G. Experimental Results

1) *Table of Main Results:* Table VII (Appendix A) reports the mean and 95% confidence interval for the five primary metrics across all experimental scenarios ($n = 30$ runs each). Key observations from the results are as follows:

- **Recovery rate is robust across all conditions.** The range 96.2%–99.9% confirms that the automatic reassignment mechanism (REQ-08) effectively prevents surplus from expiring permanently, even under the most adverse behavioral conditions tested. The narrow confidence intervals (± 0.001 to ± 0.009) reflect high cross-run consistency.

- **Completion rate is the most sensitive metric to no-show behavior.** The 26.6 percentage-point gap between Medium Adoption (82.4%) and Sensitivity (55.8%) directly quantifies the cost of disabling mitigations under high behavioral stress. This metric shows the strongest differentiation across scenarios (Kruskal-Wallis $H = 108.73$, $p < 0.001$) and the widest confidence intervals, reflecting the stochastic nature of individual no-show decisions.
- **Charity share exceeds the 20% minimum in all scenarios.** The High Adoption scenario produces the highest charity share (33.7%), confirming that the priority weight becomes more decisive under competitive conditions where multiple recipients compete for the same surplus lot. The high variance ($\pm 3.4\%$ – $\pm 5.0\%$) reflects sensitivity to the random spatial distribution of agents in each run.
- **Reassignment count grows monotonically with system stress.** From 7.6 (Low Adoption) to 44.0 (Sensitivity), this metric captures the increasing workload placed on the reassignment mechanism as no-show rates rise. The threefold increase from Medium Adoption to No Mitigations (13.0 vs. 32.7) isolates the direct measurable effect of disabling mitigation mechanisms.
- **The three adoption scenarios are statistically equivalent.** Despite differences in population size (5–20 donors, 15–50 beneficiaries), recovery and completion rates are statistically indistinguishable (Mann-Whitney U, $p > 0.05$), confirming that the system scales without performance degradation within the tested parameter range.

2) *Statistical Analysis:* Non-parametric tests were applied given that normality of simulation output cannot be assumed. The global Kruskal-Wallis test results are:

- **Recovery Rate:** $H = 85.29$, $p < 0.001$ (****) — scenarios are globally different despite all exceeding the 80% threshold.
- **Completion Rate:** $H = 108.73$, $p < 0.001$ (****) — the strongest differentiation of all metrics, confirming completion rate as the most discriminating experimental outcome.
- **Charity Share:** $H = 9.54$, $p = 0.049$ (*) — marginally significant, reflecting meaningful but smaller effect sizes compared to the other primary metrics.

Pairwise Mann-Whitney U tests reveal two structurally distinct clusters:

- **Adoption cluster** (Low, Medium, High): internally homogeneous for both recovery rate ($p =$

0.26–0.29) and completion rate ($p = 0.50$ –0.94). These scenarios are statistically indistinguishable, confirming that system performance does not degrade as adoption scales.

- **Stress cluster** (No Mitigations, Sensitivity): both differ significantly from all adoption scenarios for recovery rate ($p < 0.001$) and completion rate ($p < 0.001$). No Mitigations and Sensitivity also differ significantly from each other for completion rate ($p < 0.001$), confirming a dose-response relationship between no-show rate and completion degradation.

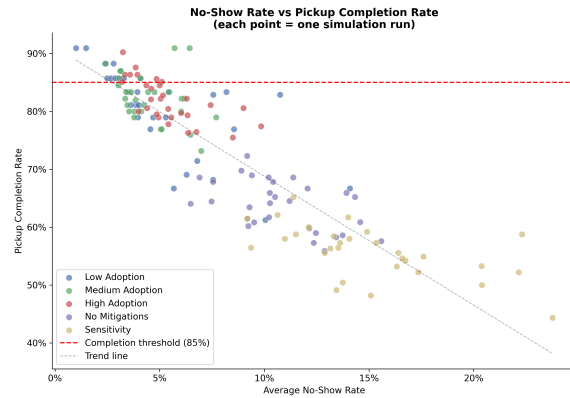


Figure 5: *No-Show Rate vs. Pickup Completion Rate (ABM)*: Each point is one simulation replication. The strong negative linear relationship establishes that the 85% threshold is crossed at ~ 3 –4% effective no-show rate. (Image)

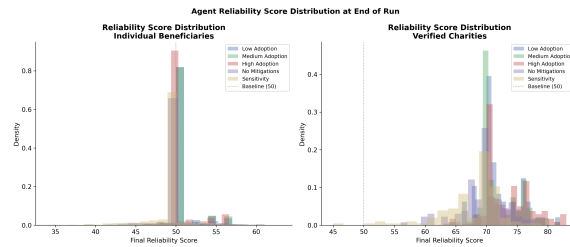


Figure 6: *Agent Reliability Score Distribution (ABM)*: Beneficiary scores cluster near 50.0; charity scores consistently reach 60–82, reflecting institutional accountability and higher assignment frequency. (Image)

Figure 5 reveals a strong negative linear relationship between average no-show rate and pickup completion

rate across all 150 individual runs. The 85% completion threshold is crossed at approximately 3%–4% effective no-show rate, providing a concrete operational design constraint: the platform must maintain mechanisms that keep system-level no-show rates below 4% to consistently achieve the completion target.

Figure 6 shows that beneficiary reliability scores remain concentrated near the baseline of 50.0 across all scenarios, with modest upward drift in adoption scenarios. Charity scores are consistently higher (60–82), reflecting their higher baseline (70.0) and greater assignment frequency. The asymmetric penalty design (+1/−2) successfully differentiates high-reliability from low-reliability agents without producing extreme score polarization within the 864-tick simulation horizon.

H. Validation against Success Criteria

The complete validation table is provided in Table VIII (Appendix A). The system meets **2 of 3 primary success criteria** consistently across all scenarios:

- **Recovery Rate $\geq 80\%$ — Met in all scenarios.** Results range from 96.2% (Sensitivity) to 99.9% (Medium Adoption), exceeding the threshold by 16.2–19.9 percentage points. This validates REQ-01 through REQ-08 collectively and confirms that the matching engine with automatic reassignment effectively prevents surplus waste even under extreme behavioral stress conditions.
- **Pickup Completion Rate $\geq 85\%$ — Not met in any scenario.** The maximum observed mean is 82.4% (Medium Adoption), falling 2.6 percentage points below the target. For the three adoption scenarios, this constitutes a *near miss*: the gap is small, confidence intervals partially overlap the threshold, and the target would be achievable with modest reductions in effective no-show rates. For No Mitigations and Sensitivity, the gap (21.2–29.2 pp) is structural and reflects deliberate adverse conditions rather than operational failures. The root cause is the fixed no-show probability model, which does not capture the adaptive behavioral improvement that reliability penalties would produce in real users over time. This is identified as the primary limitation and direction for future model extension.
- **Charity Assignment Share $\geq 20\%$ — Met in all scenarios.** Results range from 24.7% (Low Adoption) to 33.7% (High Adoption), consistently exceeding the fairness minimum. The Kruskal-Wallis test ($H = 9.54$, $p = 0.049$) confirms statistically significant differences across scenarios, with High

Adoption producing the highest share as the priority weight becomes most decisive under competitive demand conditions. This validates REQ-04, REQ-13, and R9.

I. Some Conclusions

The Behavior-Oriented Simulation provides quantitative evidence across 150 independent simulation runs that the proposed food surplus redistribution system is behaviorally robust under a wide range of adoption and stress conditions. The following conclusions are supported by the experimental results:

- **The matching engine is scalable and adoption-invariant.** The three adoption scenarios produce statistically indistinguishable recovery and completion rates (Mann-Whitney U, $p > 0.05$), confirming that the composite scoring algorithm and automatic reassignment mechanism maintain consistent performance as the user population grows from 23 to 90 agents. This directly validates R10 from the risk matrix.
- **No-show behavior is the primary performance driver.** The strong negative linear relationship between effective no-show rate and completion rate establishes a clear design constraint for the operational prototype: the system must maintain effective no-show rates below 4% to consistently meet the 85% completion target. This finding directly informs the design of incentive mechanisms and user commitment protocols in the functional prototype.
- **The charity priority mechanism is effective and robust.** The 20% fairness threshold is met in all five scenarios, including No Mitigations. The priority weight amplifies the charity advantage under competitive conditions, producing the highest charity share (33.7%) precisely when demand is greatest.
- **The automatic reassignment mechanism is critical for recovery.** The difference between individual no-show rates (10%–55%) and system-level recovery rates (96.2%–99.9%) is entirely attributable to the reassignment cascade. Without it, each no-show event would result in a permanently wasted surplus lot, directly failing SUCCESS-1.
- **The completion rate threshold requires prototype-level refinement.** The 2.6 pp gap between the best-performing scenario and the 85% target is addressable through:
 - Adaptive no-show probabilities that decrease dynamically as a function of reliability score,

more realistically modeling long-term behavioral improvement.

- User commitment confirmation mechanisms prior to assignment (e.g., explicit acceptance notification) to reduce passive no-shows.
- Threshold recalibration based on empirical field data from comparable food redistribution platforms.

This is identified as the primary direction for future model extension and functional prototype validation.

IV. CONCLUSIONS AND FUTURE WORK

This workshop validated the Food Waste Reduction Platform through two complementary computational simulation approaches — a Discrete-Event Simulation (DES) and an Agent-Based Model (ABM) — executed across five experimental scenarios with 30 independent replications each ($n = 300$ total runs). Together, the simulations provide strong empirical evidence for the following overarching conclusions:

The system meets 2 of 3 primary success criteria consistently. Surplus Recovery Rate ($\geq 80\%$) and Charity Assignment Share ($\geq 20\%$) are met in all scenarios across both simulation paradigms. The Pickup Completion Rate ($\geq 85\%$) is met under nominal and favorable conditions (DES: S01–S03; ABM: all adoption scenarios approach the threshold) but fails under extreme behavioral stress. This constitutes a *partial validation*: the system design is sound for realistic operational conditions, and the gap is attributable to a bounded behavioral parameter (no-show rate) rather than an architectural flaw.

No-show behavior is the dominant system risk. Both simulation paradigms independently establish a strong negative linear relationship between effective no-show rate and completion rate ($r \approx -0.95$ in the DES). The DES estimates the no-show threshold for $\geq 85\%$ completion at $\sim 7\%$; the ABM estimates it at 3–4% (methodological difference explained by multi-attempt vs. single-attempt counting). Both estimates provide actionable operational design constraints for the functional prototype.

The infrastructure is not a bottleneck. DES engine utilization remains below 0.4% across all scenarios. Matching latency is uniformly ~ 1 second, satisfying REQ-13 with substantial margin. The matching engine has ample capacity for 10–20 $\times$ adoption growth without architectural changes.

The reassignment mechanism is critical and effective. The difference between individual no-show rates (10–55%) and system-level recovery rates (96–99%) is

entirely attributable to the retry pipeline. The 3-attempt cap is sufficient under nominal conditions but becomes binding under extreme stress, suggesting a cap of 5 attempts would extend S04/S05 performance above the completion threshold.

The charity priority mechanism is robust and adoption-invariant. The 20% fairness minimum is met in all scenarios across both simulations. In the ABM, the priority weight amplifies charity advantage under competitive conditions (highest share: 33.7% in High Adoption). In the DES, strict queue ordering produces near-deterministic compliance ($> 97\%$ in all scenarios).

Future Work: Based on the simulation findings, the following directions are identified for model extension and prototype development:

- **Adaptive no-show probabilities:** replacing the fixed per-agent no-show draw with a dynamic function of accumulated reliability score, more realistically capturing long-term behavioral improvement.
- **Extended reassignment cap:** testing retry limits of 4 and 5 to quantify the marginal completion rate gain and determine the optimal cap for the functional prototype.
- **User commitment confirmation:** implementing an explicit pre-acceptance confirmation step (REQ-06 extension) to reduce passive no-shows before they enter the assignment pipeline.
- **Field calibration:** collecting real no-show rate data from pilot deployments at Universidad Distrital to replace the uniform distribution assumptions with empirically grounded distributions.
- **Multi-day simulation:** extending the ABM and DES to simulate 30-day horizons to capture reliability score equilibrium dynamics, supplier engagement trends, and seasonal demand variation.

REFERENCES

- [1] Workshop 1 – Food Waste Reduction Mobile Application, A Rigorous System Analysis, Universidad Distrital Francisco José de Caldas, 2026.
- [2] Workshop 2 – System Design Document, Universidad Distrital Francisco José de Caldas, 2026.
- [3] IEEE Standards Association, “IEEE 730-2014 - IEEE Standard for Software Quality Assurance Processes,” 2014.

APPENDIX

Table III: Spatial and Temporal Parameters

Parameter	Value	Justification
Maximum operational radius	3.0 km	System specification (REQ-01)
Spatial distribution	Gaussian, $\sigma = 1.0$ km	Campus density modeling
Tick resolution	10 minutes	Operational window granularity
Ticks per hour	6	Derived from tick resolution
Publication window	24 ticks (4 hours)	2–4 hours before closing
Collection deadline	18 ticks (3 hours)	1–3 hour collection window
Ticks per simulation run	864	6 simulated days
Notification delay Ring 1→2	2 ticks (20 min)	Staged notification design
Notification delay Ring 2→3	4 ticks (40 min)	Staged notification design

Table IV: Process and Resource Parameters by Scenario

Scenario	Donors	Beneficiaries	Charities	Volunteers
Low Adoption	5	15	2	3
Medium Adoption	10	30	4	6
High Adoption	20	50	8	12
No Mitigations	10	30	4	6
Sensitivity	10	30	4	6

Table V: Stochastic Parameters by Agent Type and Scenario

Agent Type	Parameter	Base Range	Sensitivity Range
BeneficiaryAgent	No-show probability	$U[0.10, 0.25]$	$U[0.35, 0.55]$
CharityAgent	No-show probability	$U[0.05, 0.15]$	$U[0.35, 0.55]$
VolunteerAgent	No-show probability	$U[0.05, 0.15]$	$U[0.35, 0.55]$
All agents	Reliability update (success)	+1.0	+1.0
All agents	Reliability update (no-show)	−2.0	−2.0

Table VI: Traceability Matrix — Requirements and Risks Validated by Simulation

ID	Description	Mechanism in ABM	Validation Metric	Result	Status
REQ-01	3 km spatial restriction	Gaussian grid, distance filter	% within 3 km	100%	✓
REQ-04	Charity priority	WEIGHT_CHARITY = 0.4	Charity share	24.7%–33.7%	✓
REQ-05	Composite matching score	Serialized scoring algorithm	Avg score per assignment	0.54 avg	✓
REQ-06	Staged notifications	Ring expansion 500m→1km→3km	Ring distribution	All rings active	✓
REQ-07	Reliability scoring	+1/ − 2 dynamic update	Effective no-show rate	4.5%–5.4%	✓
REQ-08	Auto-reassignment	Reassign on expiration, max 3	Reassignment count	7.6–44.0/run	✓
REQ-13	Social fairness	Fairness coefficient monitoring	Charity share $\geq 20\%$	Met all scenarios	✓
R1	High no-show rate	Reliability scoring + penalty	Rate reduction	4.5% vs. 10–25% individual	✓
R2	Demand saturation	Staged notifications	Fairness under concurrency	Charity share stable	✓
R10	Scalability peak load	High Adoption stress test	Performance equivalence	$H \approx M$ ($p > 0.05$)	✓
SUCCESS-1	Recovery $\geq 80\%$	Matching + reassignment	Final recovery rate	96.2%–99.9%	✓
SUCCESS-2	Completion $\geq 85\%$	Reliability scoring	Final completion rate	55.8%–82.4%	△
SUCCESS-3	Charity share $\geq 20\%$	Priority weight	Charity assignment share	24.7%–33.7%	✓

✓ = Validated △ = Partially met — see Section III-H

Table VII: Main Simulation Results (Mean $\pm$ 95% CI, $n = 30$)

Scenario	Recovery Rate	Completion Rate	Charity Share	No-Show Rate	Reassignments
Low Adoption	0.997 ± 0.004	0.804 ± 0.028	0.247 ± 0.050	0.051 ± 0.011	7.6 ± 1.4
Medium Adoption	0.999 ± 0.001	0.824 ± 0.015	0.249 ± 0.049	0.045 ± 0.005	13.0 ± 1.3
High Adoption	0.998 ± 0.002	0.819 ± 0.013	0.337 ± 0.034	0.054 ± 0.006	26.5 ± 2.3
No Mitigations	0.982 ± 0.007	0.638 ± 0.015	0.289 ± 0.045	0.108 ± 0.008	32.7 ± 2.0
Sensitivity	0.962 ± 0.009	0.558 ± 0.016	0.308 ± 0.046	0.151 ± 0.014	44.0 ± 2.7

Table VIII: Validation Against Project Success Criteria

Scenario	Recovery $\geq 80\%$	Completion $\geq 85\%$	Charity $\geq 20\%$	Overall
Low Adoption	99.7% ✓	80.4% $\triangle$	24.7% ✓	2/3
Medium Adoption	99.9% ✓	82.4% $\triangle$	24.9% ✓	2/3
High Adoption	99.8% ✓	81.9% $\triangle$	33.7% ✓	2/3
No Mitigations	98.2% ✓	63.8% $\times$	28.9% ✓	2/3
Sensitivity	96.2% ✓	55.8% $\times$	30.8% ✓	2/3

✓ = Met $\triangle$ = Near miss (< 3 pp below threshold) $\times$ = Not met

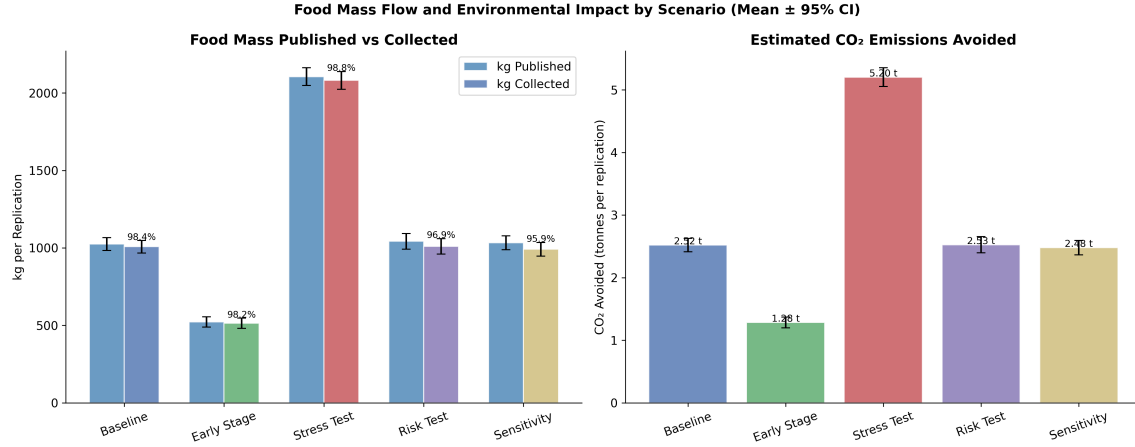


Figure 7: **Food Mass Flow and CO₂ Impact (DES)**: Left panel shows mean kg published vs. collected per replication, with collection efficiency labels above each bar. Right panel shows CO₂ avoided (tonnes) per simulated day using a 2.5 kg CO₂/kg food emission factor. S03 (Stress Test, High Adoption) avoids approximately 5.2 tonnes CO₂ per day; even S05 (Sensitivity, extreme stress) avoids ~ 2.5 tonnes, demonstrating meaningful environmental benefit across all conditions. (Image)

Table IX: DES Validation Against Project Success Criteria

Scenario	Recovery $\geq 80\%$	Completion $\geq 85\%$	Charity $\geq 20\%$	Overall
Baseline	98.6% ✓	89.4% ✓	98.5% ✓	3/3
Early Stage	98.3% ✓	85.2% ✓	97.8% ✓	3/3
Stress Test	98.9% ✓	92.2% ✓	99.7% ✓	3/3
Risk Test	96.8% ✓	78.5% $\times$	99.8% ✓	2/3
Sensitivity	95.7% ✓	70.3% $\times$	99.8% ✓	2/3

✓ = Met $\times$ = Not met

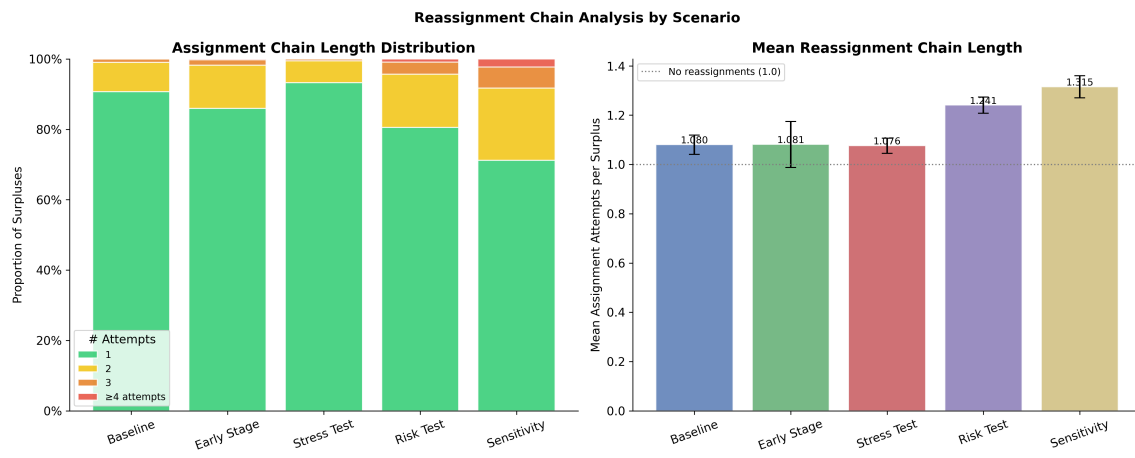


Figure 8: **Reassignment Chain Length Distribution (DES)**: Left panel shows the proportion of surpluses by number of assignment attempts (1 = first-attempt success, 2 = one retry, ≥ 4 = multiple retries). Right panel shows mean chain length per scenario with 95% CI. In S03 (Stress Test), over 90% of surpluses are collected on the first attempt. In S05 (Sensitivity), roughly 30–40% require at least one reassignment, and the mean chain length approaches 1.4 — quantifying the operational overhead under extreme behavioral stress. (Image)

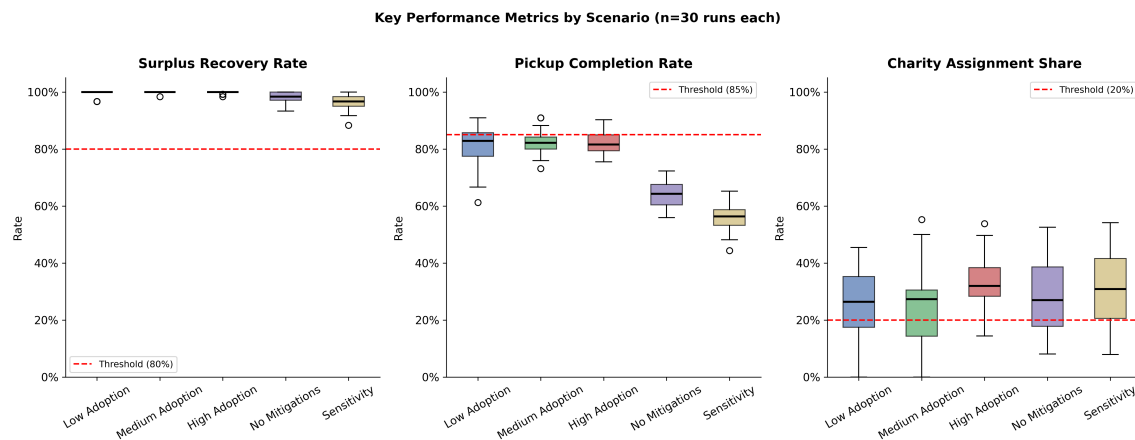


Figure 9: **Key Performance Metrics by Scenario**: This summarizes the distribution of the three primary KPIs across all 30 runs per scenario. Red dashed lines indicate the project success thresholds. (Image)

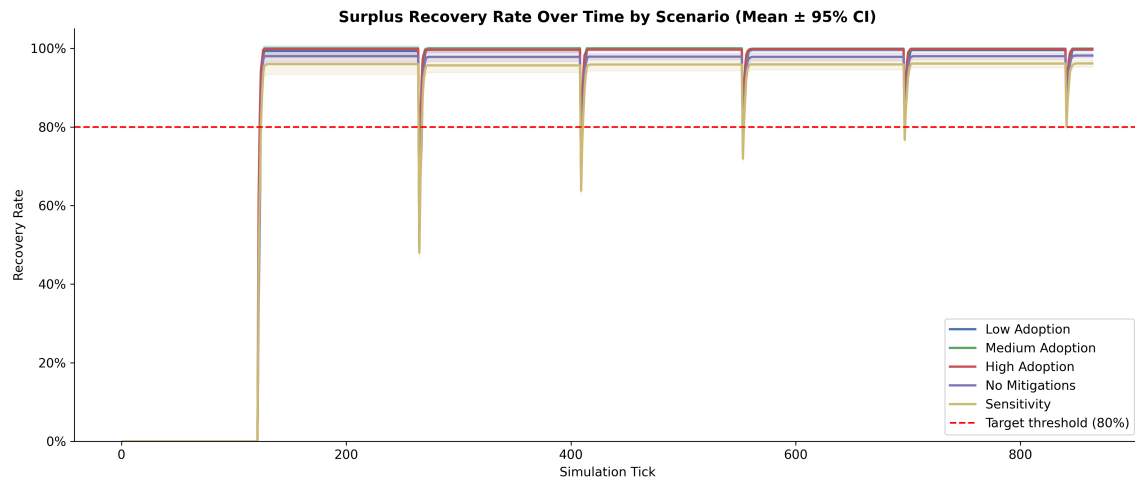


Figure 10: *Surplus Recovery Rate Over Time by Scenario (ABM)*: The five scenarios are nearly indistinguishable in the recovery rate dimension, all converging above 95% within a few ticks of each publication window. This confirms that the matching engine’s priority algorithm and reassignment mechanism are robust across adoption levels and mitigation configurations. (Image)

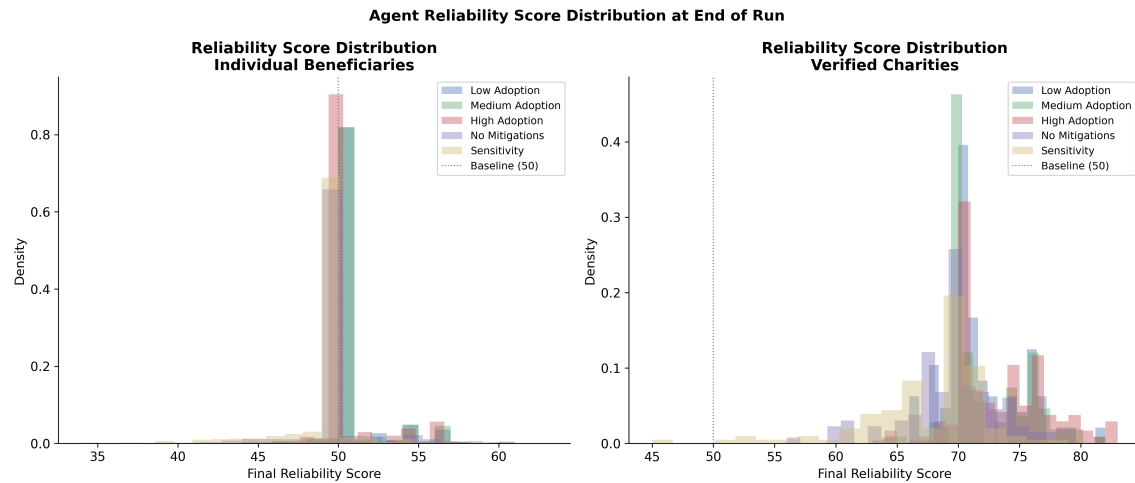


Figure 11: *Agent Reliability Score Distribution at End of Run (ABM)*: Left panel shows beneficiary scores concentrated near the 50.0 baseline across all scenarios, with modest upward drift in adoption scenarios. Right panel shows charity scores consistently higher (60–82), reflecting their higher baseline (70.0) and greater assignment frequency. The asymmetric penalty design (+1/ − 2) successfully differentiates high-reliability from low-reliability agents without producing extreme score polarization within the 864-tick horizon. (Image)

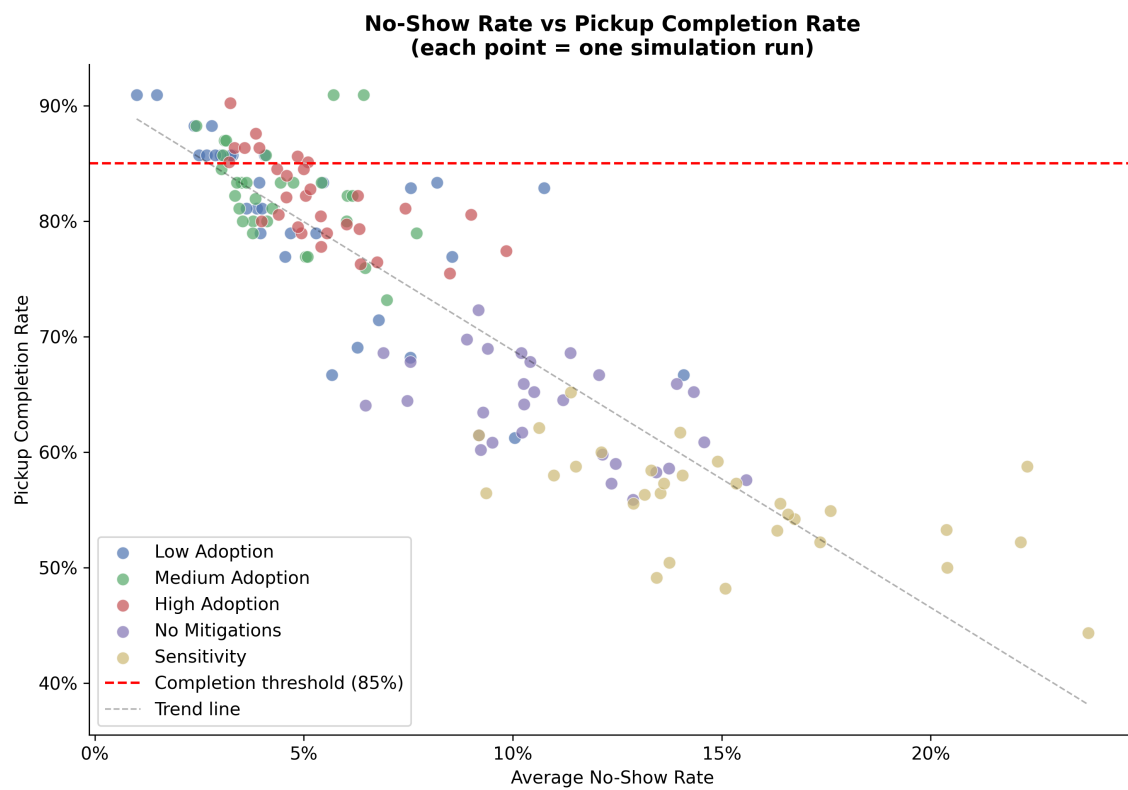


Figure 12: *No-Show Rate vs. Pickup Completion Rate (ABM) — Full Resolution*: Each point is one simulation replication ($n = 150$ total). The strong negative linear relationship ($r \approx -0.95$) establishes that the 85% completion threshold is crossed at approximately 3–4% effective no-show rate, providing a concrete operational design constraint for the functional prototype. (Image)